Cardiovascular & Peripheral Vascular Assessment

NURA 330 Health Assessment | Module 5 Lecture Study Guide

Exam 2 content · reviewed for the final

Lecture roadmap: Case: Eleanor → Heart location and anatomy → Blood flow, valves, systole/diastole → Conduction system → Lifespan → Subjective interview and A&P connections → Chest pain (PQRST and priority actions) → Heart attack symptoms by gender → Pediatric chest pain → SDOH → Vital signs → Inspection (skin, JVD, precordium, extremities) → CWMS → Pulses → Capillary refill → Edema → Heart sounds and APE To Man → Abnormal sounds → Findings to report → Diagnostics → Case wrap-up → Documentation

Exam 2 study guide topics: heart location and anatomy; conduction system; lifespan; subjective interview and PQRST; A&P links to symptoms; MI symptoms by gender; pediatric chest pain; SDOH; vital signs and perfusion; **JVD**; extremities; **pulses and capillary refill**; **edema grading**; **auscultation landmarks (APETM)**; **S1 and S2**; auscultation tips; findings to report; diagnostics and terminology.

How to read this guide: Everything without a tag comes from the lecture slides or the slide notes. Anything else is tagged: STUDY GUIDE emphasized on the exam study guide but not spelled out on the slides; ADDED information not in the slides; CLARIFIED a slide point explained further; CORRECTED an error or inconsistency in the slides. Every tag is also listed in the notes at the end.

Case: Eleanor, 72. Admitted for increasing fatigue and shortness of breath with activity: "Yesterday, I had to stop halfway up the stairs because I was so short of breath." **HR 104 | BP 154/88 | RR 22 | SpO_2 94% RA | T 37 °C**. One abnormal number may not mean much; **the pattern matters**. Cardiovascular problems often show up as **changes in function**, so ask what she can do now compared with before.

Anatomy and Physiology

Location of the Heart

- Sits in the **mediastinum**, behind the sternum, between the lungs, resting on the diaphragm.

- **Base** (top, where the great vessels attach): faces up and slightly right, around the **2nd intercostal space (ICS)**.
- **Apex** (pointed bottom): tilts down and left, at the **5th ICS, midclavicular line**. This is where you find the **point of maximal impulse (PMI)** and the **apical pulse**.
- The heart sits right next to the lungs and trachea, which is why respiratory problems affect the heart and vice versa.

Layers, Chambers, Vessels, Valves

Layers	**Pericardium:** protective sac with a little serous fluid to reduce friction. **Myocardium:** thick muscle that contracts. **Endocardium:** smooth inner lining of chambers and valves
Chambers	**Atria receive** blood; **ventricles pump**. Right atrium receives deoxygenated blood from the body; right ventricle pumps to the lungs; left atrium receives oxygenated blood from the lungs; **left ventricle pumps to the body** and has the **thickest wall**
Vessels	Venae cavae → right atrium. Pulmonary artery: RV → lungs (deoxygenated; still an artery because it carries blood **away** from the heart). Pulmonary veins: lungs → LA. Aorta: LV → body
Valves	One-way doors. **AV valves:** tricuspid (RA→RV), mitral (LA→LV). **Semilunar valves:** pulmonic (RV→pulmonary artery), aortic (LV→aorta)

Body → vena cava → RA → tricuspid → RV → pulmonic → pulmonary artery → lungs → pulmonary veins → LA → mitral → LV → aortic → aorta → body

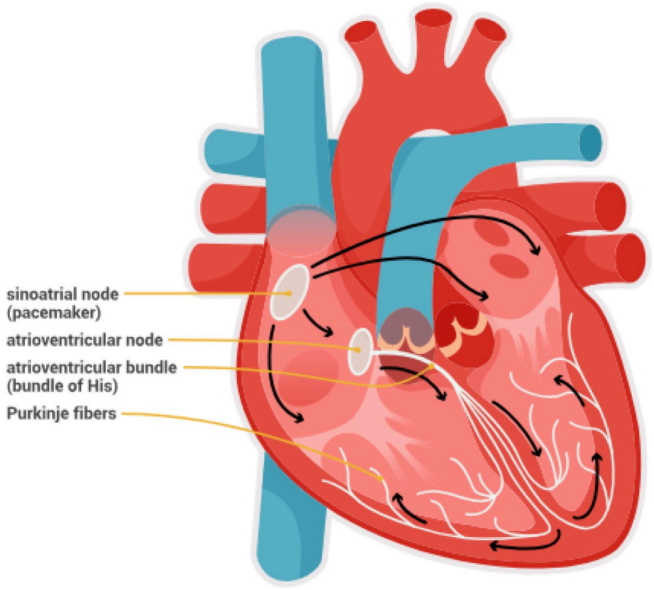
- **Diastole:** ventricles relax and fill. **Systole:** ventricles contract and eject.
- **S1** ("lub") = **mitral and tricuspid close** (start of systole). **S2** ("dub") = **aortic and pulmonic close** (start of diastole).
- **Vascular system:** arteries (thick, elastic, high pressure) → arterioles (control resistance) → capillaries (exchange) → veins (low pressure, blood reservoir).
- **Venous return** is helped by venous valves, skeletal muscle contraction, and breathing pressure changes.

The two questions of every CV assessment: Is the heart pumping effectively? and **Is blood reaching the tissues and returning appropriately?** That is why we check HR and rhythm, heart sounds, BP, pulses, cap refill, skin color and temperature, edema, and activity tolerance.

Electrical Conduction System

SA node → AV node → Bundle of His → Bundle branches → Purkinje fibers

- The **SA node** (right atrium) is the natural **pacemaker**. The impulse spreads across the atria, which contract.
- The **AV node briefly delays** the impulse so the ventricles can fill.
- The impulse then travels down the bundle of His, bundle branches, and Purkinje fibers, so the ventricles contract together.
- Abnormal impulses make the heart beat too fast, too slow, or irregularly. Assess for **dizziness, palpitations, chest discomfort, SOB, weakness, syncope**, and check that **apical and peripheral pulses match**.
- The electrical system tells the heart *when* to contract; the muscle provides the *force*.



Cardiac conduction system (lecture slide)

Lifespan Considerations

Infants & children	Pregnancy	Older adults
Faster HR than adults; use age-specific HR and BP norms; assess feeding/activity tolerance and perfusion. **S3 can be normal**; about half of children have (often innocent) murmurs	↑ blood volume, HR, and cardiac output; a mild HR increase may be expected; monitor BP, edema, dyspnea	Stiffer arteries → ↑ **systolic BP**; assess for **orthostatic hypotension**; watch for irregular rhythms, edema, decreased activity tolerance

Interpret findings by age, baseline function, and symptoms. You cannot memorize one set of normals for everyone.

Subjective Assessment

Area	Ask about
Symptoms	Chest pain/pressure, shortness of breath, palpitations, dizziness/syncope, swelling, fatigue/activity intolerance (use OLDCARTS)
Past history	HTN, CAD, heart failure, valve disease, PVD, high cholesterol; prior MI or stroke; procedures (stent, ablation, pacemaker, surgery)
Medications	BP, cholesterol, rhythm, chest pain meds, diuretics, anticoagulants; herbs/supplements
Family history	Heart disease, stroke, HTN, **premature** cardiovascular disease
Lifestyle	Smoking/vaping; physical activity (type, frequency, duration); diet; alcohol and substances

A&P Connections to Symptoms

Think about	Ask about	Connection
Coronary blood flow	Chest pain/pressure, radiation	↓ blood flow to heart muscle → ↓ O_2 → ischemic pain

Think about	Ask about	Connection
Heart–lung circulation	Dyspnea, orthopnea, cough, activity tolerance	Poor pumping backs fluid into the lungs
Electrical system	Palpitations, dizziness, syncope	Abnormal rate/rhythm can reduce cardiac output
Venous return / fluid	Edema, weight gain, nocturia	Venous congestion → fluid buildup
Peripheral perfusion	Leg pain, color/temperature changes, wounds	Arteries supply O_2; veins return blood

In the ROS, follow up pertinent positives with OLDCARTS; document negatives as "denies ..."

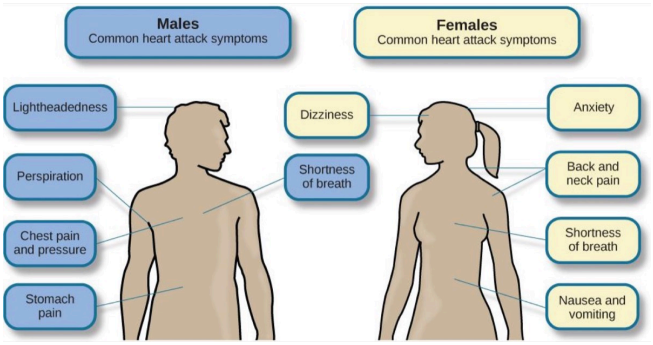
Chest Pain: Priority Approach

Phase	Actions
Subjective (PQRST)	Location and quality (point to it; sharp, dull, **crushing**?); onset and **radiation to arm, jaw, neck, back**; triggers (exertion vs. rest) and relief; associated symptoms: **nausea, diaphoresis, dyspnea, numbness/tingling**
Objective	Continuous VS and SpO_2; skin color (pallor, cyanosis); breathing effort; **JVD and edema** (fluid overload)
Interventions	**Never leave the patient alone.** Stop activity; **semi-Fowler's** position. Oxygen; IV access; continuous ECG; **call the rapid response team** and notify the provider

CLARIFIED The slide says to give oxygen by non-rebreather. Current chest pain guidelines give oxygen when the patient is hypoxemic (SpO_2 below about 90%) or in distress. On an exam, follow the order or protocol in the question.

Heart Attack Symptoms: Gender Differences

More typical in men	More common in women
Chest pain and pressure, sweating, lightheadedness, shortness of breath, stomach pain	Often **no classic chest pain**: **back and neck pain**, **nausea and vomiting**, shortness of breath, dizziness, **anxiety**, fatigue



Common heart attack symptoms in males and females (lecture slide)

Chest Pain in Children

- Usually **idiopathic** (benign, no organic cause found).
- **Musculoskeletal (most common):** precordial catch syndrome (sudden, sharp, brief, harmless), costochondritis.
- Respiratory (asthma, infections), GERD (burning), anxiety and stress.
- **Cardiac causes are rare.**
- **Red flags:** chest pain **during physical activity**, persistent pain, or pain with **fainting**; needs prompt evaluation.

Psychosocial and SDOH

Access to care (meds, appointments, transportation, cost); food access and usual diet; exercise, mobility, and safe spaces; stressors, coping, support; job demands, caregiving, finances; tobacco, vaping, alcohol, stimulants.

Objective Assessment

Order: **vital signs** → **inspect** → **palpate** → **auscultate**. CV assessment overlaps with respiratory and skin assessment because heart problems show up as skin and lung findings.

Vital Signs: Perfusion and Oxygenation

Measure	Expected	Abnormal / meaning
Heart rate	60–100 bpm	**Tachycardia >100** (compensation, meds, arrhythmia, anxiety). **Bradycardia <60** (may not perfuse organs)

Measure	Expected	Abnormal / meaning
Blood pressure	<120/80	**Hypotension: systolic <100** (not enough pressure to perfuse organs). **Hypertension**: slide says systolic >120 or diastolic >80 CORRECTED see note 1; high resistance makes the heart overwork. **Narrow pulse pressure** (e.g., 90/70) suggests low stroke volume
SpO_2	95–100% RA	**Hypoxemia <95%**. No reading or poor waveform may mean poor peripheral perfusion or cold hands
Respirations	12–20/min	**Tachypnea >20** (compensation for hypoxia or poor perfusion). **Orthopnea** = SOB lying flat, a sign of fluid backing into the lungs (left-sided HF)

Compensation patterns: Low SpO_2 → tachypnea. Low BP → tachycardia. **Red-flag pattern: tachycardia + hypotension + tachypnea + low SpO_2** = systemic cardiovascular failure and tissue hypoxia.

Inspection

Area	What to look for
Skin color & perfusion	Face, lips, fingertips. **Cyanosis** (blue: low oxygenation/perfusion); **pallor** (paleness: reduced blood flow, oxygen, or RBCs). In darker skin, check palms, conjunctiva, inner lower lip
Jugular vein distention	Reflects **right atrial pressure**. Most visible on the **right side of the neck**. **Should NOT be present upright or with HOB at 30–45°**
Precordium	Chest over the heart: deformities, surgical scars, abnormal pulsations or heaves
Upper extremities	Both arms at once for symmetry; pallor, cyanosis, redness; skin breakdown or lesions

Area	What to look for
Lower extremities	**Hair distribution**, ulcers (size and location), **edema**, distended superficial veins; **compare calves** for DVT signs

JVD Assessment

- Position the patient **supine with HOB at 30–45°**; remove the pillow to avoid flexing the neck; turn the head **slightly away** from the side you are assessing.
- Do not confuse the **carotid pulse** with the internal jugular vein.
- **Causes of JVD: right-sided heart failure**, fluid overload (kidney or liver disease), constrictive pericarditis, cardiac tamponade, pulmonary hypertension.
- JVD is painless but a red flag, **especially with shortness of breath or leg swelling**.

ADDED **Telling the carotid from the jugular:** the carotid pulse is a **single, strong outward beat you can feel**; it does not change with position. Jugular pulsations are **soft, have multiple waves, are not palpable**, and drop when the patient sits up.

Arterial vs. Venous Problems in the Legs

	Arterial insufficiency (PAD)	Venous insufficiency
Hair	**Loss of hair; shiny, hairless skin** (follicles need arterial blood)	Normal hair distribution
Skin	**Pale, cool**	**Brown discoloration** (hemosiderin staining), varicose veins
Pulses / refill	**Weak or absent pulses; slow cap refill**	Pulses usually present
Edema	Usually minimal	**Chronic swelling**

DVT red flag: **Unilateral** calf redness, warmth, tenderness, or swelling. Compare both legs; report right away.

CWMS: Color, Warmth, Movement, Sensation

Color (inspect)	Pallor, cyanosis, or rubor (dusky redness)

Warmth (palpate)	Compare both sides with the **back of your hand**. **Cool** may mean arterial insufficiency; **warm** may mean inflammation, venous stasis, or DVT
Movement	Wiggle fingers and toes; flex and extend (motor nerves and muscle perfusion)
Sensation	Light touch on fingers and toes, both sides equal? (detects neuropathy or nerve compression)

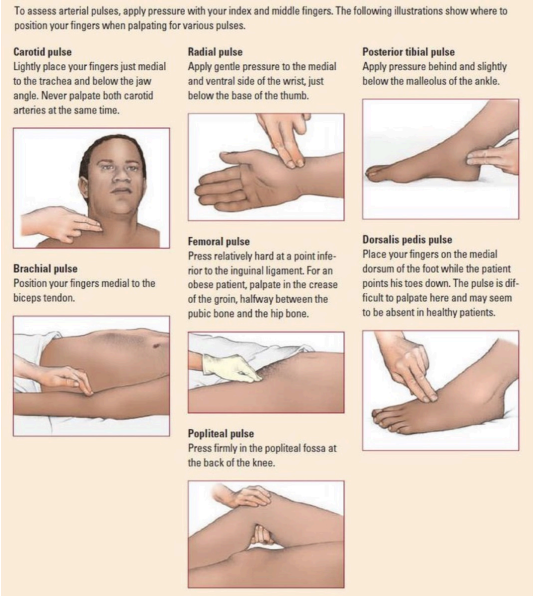
Pulses, Capillary Refill, and Edema

Pulse Grading (slide scale)

Grade	Meaning
0	Absent, cannot be felt
1+	Weak, thready, diminished
2+	**Normal, brisk (expected)**
3+	Increased, strong
4+	Bounding, full

CORRECTED The slide uses a 0–4+ scale; its speaker notes describe a 0–3 scale (3 = bounding). Both are used in practice; see note 2.

- **Compare both sides** for rate, rhythm, and quality: radial, brachial, posterior tibial, dorsalis pedis.
- **Carotid: one side at a time**, never both together (it can reduce blood flow to the brain).
- **Posterior tibial:** just behind and below the **medial malleolus** (inner ankle bone).
- **Dorsalis pedis:** top of the foot, just lateral to the big toe's extensor tendon.
- **Cannot find a pulse?** Decide whether it is new or chronic; use a **Doppler**. If absent even with Doppler, it may be an emergency: notify the provider.
- Document **location, strength (grade), symmetry**, and rate/rhythm.



Where to palpate peripheral pulses (lecture slide)

Capillary Refill

Press a fingernail or toenail until it blanches (turns white), release, and time how long color takes to return. With good blood flow, color returns within **2–3 seconds**. Longer than that = poor perfusion.

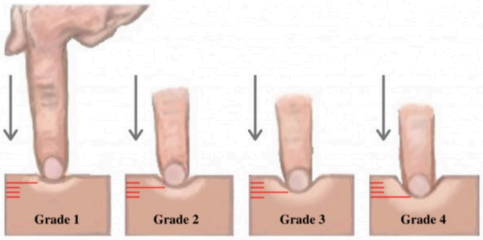
CORRECTED The Mod 4 slides use "<2 seconds" and the Mod 5 slides use "2–3 seconds." See note 3.

Edema

Fluid retention leaks into dependent tissues (third spacing), so lower legs swell first. **The higher the edema goes up the leg, the more fluid overload.** High sodium intake also causes fluid retention. Press firmly for a few seconds over the shin or ankle and observe the pit.

Grade	Pit depth	Rebound (slide)
1+	0–2 mm	Immediately
2+	3–4 mm	<15 seconds
3+	5–6 mm	Up to 30 seconds
4+	8 mm	>20 seconds CORRECTED see note 4

Grade 1	0–2 mm indentation; rebounds immediately.
Grade 2	3–4 mm indentation; rebounds in < 15 seconds.
Grade 3	5–6 mm indentation; up to 30 seconds to rebound.
Grade 4	8 mm indentation; > 20 seconds to rebound.

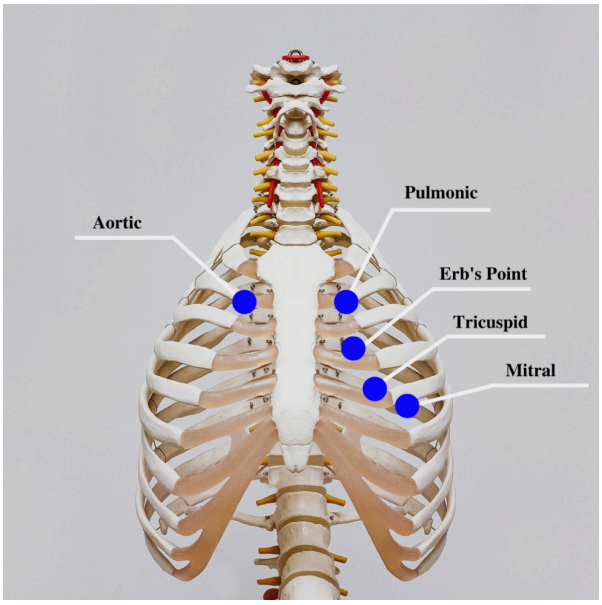


Pitting edema grading scale (lecture slide)

Auscultation

Landmarks: "APE To Man"

Area	Location
A: Aortic	**2nd ICS, right** sternal border
P: Pulmonic	**2nd ICS, left** sternal border
E: Erb's point	**3rd ICS, left** sternal border
T: Tricuspid	**4th ICS, left** sternal border
M: Mitral (apical)	**5th ICS, left midclavicular line** (PMI / apical pulse)



Cardiac auscultation points (lecture slide)

Technique

1. Start at the **aortic** area with the **diaphragm**; move through P, E, T, M.
2. Identify S1 and S2; assess **rate and rhythm**; listen for extra sounds.
3. Repeat with the **bell** (for low-pitched sounds).
4. For the mitral area in a patient with breasts, ask them to lift the breast tissue so you are on the chest wall.
5. **Apical pulse:** 5th ICS, midclavicular line; **count for a full 60 seconds**.

Tips: have the patient **lean forward or lie on the left side**; if lung sounds interfere, ask them to **hold their breath briefly** (≤10 seconds); turn off the TV and close the door; explain ("Please try not to talk while I listen"); practice.

Expected Heart Sounds

	S1 ("lub")	S2 ("dub")
Caused by	**Mitral and tricuspid** (AV valves) closing	**Aortic and pulmonic** (semilunar valves) closing
Marks	**Start of systole**	**Start of diastole**
Pitch	Dull, **lower-pitched**, longer CORRECTED see note 5	**Higher-pitched**, shorter
Loudest at	**Apex** (mitral area)	**Base** (aortic and pulmonic areas): S2 louder than S1 there

Systole = the silent pause between S1 and S2. **Diastole** = the pause between S2 and the next S1 (normally longer).

ADDED **Tip:** If you are unsure which sound is S1, feel the carotid pulse while you listen. **S1 happens at the same time as the carotid pulse.**

Unexpected Heart Sounds

Sound	Description
S3 ("Ken-TUCK-y")	Low-pitched, early diastole, right after S2. From rapid ventricular filling or **volume overload**. **Normal in children**; in older adults with edema, weight gain, and orthopnea it suggests heart failure
S4 ("TEN-nes-see")	Low-pitched, late diastole, right before S1. From a **stiff, noncompliant ventricle**. **Abnormal**
Murmurs	Blowing or whooshing sound from **turbulent blood flow** through valves or defects. Innocent or pathologic. About 50% of children have murmurs (e.g., Still's murmur, ages 2–6). **A new murmur must be reported**
Clicks	High-pitched, just after S1, at maximal valve opening (e.g., mitral valve prolapse)
Pericardial friction rub	Grating or scratching from an inflamed pericardium

Extra sounds are hard for beginners to distinguish. **If you hear anything new or different, notify the provider or an experienced clinician.** Use the **bell** for S3, S4, and murmurs.

Abnormal Findings to Report

Category	Report
Weight / fluid	Rapid unexplained weight gain; new or worsening edema
Skin / perfusion	Cool, overly warm, or diaphoretic skin; cyanosis or pallor; **cap refill >2–3 sec**; decreased or unequal hair distribution
Pulses	Absent (even with Doppler), weak/thready, or bounding; new irregular pulse

Category	Report
Rate & rhythm	Symptomatic tachycardia at rest (>100) or bradycardia (<60); new irregular rhythm; **new extra heart sounds** (murmur, S3, S4)
Blood pressure	New systolic **<100**; orthostatic changes
Symptoms	Chest pain, calf pain, worsening SOB; impaired movement or sensation of fingers or toes
Vascular	**Unilateral warmth, redness, tenderness, or edema (possible DVT)**; **JVD upright or at HOB 30–45°**

Nursing interventions: maintain oxygenation, cardiac output, and perfusion; hydration and nutrition; pain relief, comfort, rest; give meds and assess response; prevent complications; emotional support.

Diagnostic Tests and Labs

Electrocardiogram (ECG)	Electrical activity
Echocardiogram	Structures and blood flow (ultrasound)
Heart catheterization	Evaluates the heart (and lungs)
CT / cardiac MRI	Detailed images of heart and vessels; MRI shows the moving heart
Labs	**CBC, ABG, BNP** (heart failure), **troponin** (heart muscle damage) ADDED, electrolytes

Back to Eleanor

New findings: exertional SOB; **3-pillow orthopnea**; **5-lb weight gain in 1 week**; denies chest pain. HR 104 regular; S1, S2 present; **S3 at the apex**; BP 154/88. Radial pulses 2+ bilaterally; **dorsalis pedis 1+ bilaterally**; **cap refill ~3 sec in toes**; **2+ bilateral ankle edema**.

Expected	Unexpected
Alert, oriented; no chest pain; S1, S2 present; radial pulses 2+ and equal; extremities warm and similar in color; no one-sided redness or tenderness	HR 104; BP 154/88; exertional dyspnea; 3-pillow orthopnea; 5-lb weight gain; S3; 2+ edema; diminished pedal pulses; slower cap refill

What it suggests: the heart may not be pumping well enough to meet demand, and fluid is accumulating (edema, weight gain, orthopnea, S3). **Prioritize:** worsening SOB, orthopnea, rapid weight gain, new or worse edema, tachycardia, the S3 in context.

Documentation

Be specific: S1, S2, any murmur or extra sound; exact HR, BP, edema grade and location; signs of circulatory distress (orthopnea, JVD).

Avoid	Instead
"Heart sounds normal"	"S1, S2 present, regular rate and rhythm, no murmurs"
"BP high" / "Heart rate elevated"	"BP 154/88" / "HR 104, regular"
"Mild edema" / "No swelling"	"2+ pitting edema bilateral ankles" / "No edema bilateral lower extremities"
"Occasional dizziness"	Describe when, how often, and what the patient was doing

Key terms: S1/S2, murmur, regular rate and rhythm (RRR), bradycardia, tachycardia, edema, orthopnea, dyspnea on exertion, JVD, auscultation, angina.

Quick Review

Topic	Key points
Apex / PMI	5th ICS, left MCL; count apical pulse 60 sec

Topic	Key points
Conduction	SA → AV (delay) → His → bundle branches → Purkinje
S1 / S2	S1 = mitral + tricuspid close, apex, start systole. S2 = aortic + pulmonic close, base, start diastole
APE To Man	2R, 2L, 3L, 4L, 5L MCL
S3 / S4	S3 Kentucky (volume overload; normal in kids). S4 Tennessee (stiff ventricle; abnormal)
JVD	HOB 30–45°, right side; present = right-sided HF / fluid overload
Pulses	0 absent, 1+ weak, 2+ normal, 3+ increased, 4+ bounding; carotid one side at a time
Cap refill	≤2–3 sec
Arterial vs. venous	Arterial: hairless, pale, cool, weak pulses. Venous: brown staining, edema, varicosities
MI in women	Back/neck pain, nausea, SOB, dizziness, anxiety; may lack chest pain
Pediatric chest pain	Usually benign/musculoskeletal; red flag = with exercise or fainting

Self-Check ADDED

1. Where is S2 loudest? *Answer: At the base (aortic and pulmonic areas).*
2. Where do you place the stethoscope to assess the tricuspid area? *Answer: 4th ICS, left sternal border.*
3. A patient's left calf is warm, red, and swollen; the right is normal. Priority? *Answer: Possible DVT; report immediately.*
4. A 55-year-old woman reports nausea, jaw and back pain, and fatigue, but no chest pain. What should the nurse consider? *Answer: A possible MI; women often present without classic chest pain.*
5. At what position should JVD be assessed? *Answer: HOB 30–45°, head turned slightly away.*
6. A low-pitched sound right after S2 in a 74-year-old with edema suggests what? *Answer: S3, associated with volume overload/heart failure.*

Notes on Sources, Additions & Clarifications

Corrections and inconsistencies in the slides:

1. **Hypertension cutoff.** The vital signs slide lists hypertension as "systolic >120 or diastolic >80." Under current ACC/AHA categories, 120–129 systolic with diastolic <80 is "elevated," and hypertension starts at **130/80**. The slide's "<120/80 = expected" is correct. Know the slide's number for this course, and the formal categories for NCLEX.
2. **Pulse scale.** The slide uses 0–4+; the speaker notes describe 0–3. Use the scale your agency uses; this guide follows the slide.
3. **Capillary refill.** Mod 4 says <2 seconds; Mod 5 says 2–3 seconds (and lists ">2–3 seconds" as reportable). Both appear in references. Treat ≤2 seconds as clearly normal and >3 seconds as clearly abnormal.
4. **Edema scale.** The slide's table says 4+ rebounds in ">20 seconds," which is less than 3+'s "up to 30 seconds." That is likely a typo. Many references give longer rebound times for 3+ and 4+ (up to several minutes). Focus on **pit depth**, which is consistent, and ask your instructor which times they want.
5. **S1 pitch.** One slide lists S1 and S2 both as "high"; other slides and the notes say S1 is dull and **lower-pitched**, and S2 is higher-pitched. This guide uses the latter.
6. The slide notes refer to "pleural friction rubs" in the heart sounds section; in the heart it is a **pericardial** friction rub (as the slide itself says).

From the Exam 2 study guide: the topic list in the blueprint box.

Added information (not in the slides):

- Telling the carotid pulse from the jugular vein.
- Feeling the carotid pulse to identify S1.
- What troponin measures.
- The arterial vs. venous table is organized from the slide notes; the "minimal edema" row for arterial disease is added.
- The self-check questions.

Clarified: oxygen for chest pain is given based on hypoxemia under current guidelines; the slide says non-rebreather.